CIA - 1

Course: Mobile computing
Code : ITA0329

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1) Configuration of cellular network performance

Given:

Hexagonal cellular network = 1,200 km^2

cell radius = 1.8 KM

Available spectrum = 48 MHz

Full-duplex channel = 200 KHz

Cluster sizes (N) = 7, 4

Reduction in channel usage = 15%

i) calculate number of cells:

radius (R) = 1.8

$A_c = \frac{3\sqrt{3}}{2} R^2$

$= 2.598 (1.8)^2$

$= 8.5199\ km^2$

$= \frac{1200}{8.5199}\ km^2 = 143$ cells

∴ Approximate no of cells = 143

ii) Total available channels:

$C = \frac{\text{Available spectrum}}{\text{full duplex channel}}$

$= \frac{48 \times 1000}{200}$

$= 240$ channels

Considering 15% channel reduction

$= 240 \times \frac{85}{100}$

$= 204$ channels

iii) Theoritical and effective channel per cell

when

N = 7: = Total channel / N
= 240 / 7
= 34.29

N = 4: = Total channel / N
= 240 / 4
= 60

Effective channel per cell

= 34.29 × 0.85
= 29 channel / cell

Effective channel
= 204 / 4
= 60 × 0.85
= 51 channel / cell

iv) Frequency reuse Ratio and distance

for N = 7

$Q = \sqrt{3(N)}$

$= \sqrt{21}$

$Q = 4.58$

$D = Q \times R$
$= 4.58 \times 8$
$D = 8.25$ km

for N = 4

$Q = \sqrt{3(4)}$

$= \sqrt{12}$

$= 3.46$

$D = Q \times R$
$= 3.46 \times 8$
$D = 6.24$ km

Therefore, N = 7 is more suitable choice

2) Markoff delay and Roaming performance:

Given:-

Cell boundary crossing = 15

BSS coordination delay = 120 ms

Inter PLMN transcitions = 2

Authentication delay = 350 ms

Registration delay = 250 ms

failed handoffs = 3

Recovery delay = 400 ms

Data leak = 512 Kbps

Maximum permitted interuption = 35

i) Normal Delay per handoff:

= 350 + 250

= 600

= 200

ii) Total normal handoff delay

= Number of hand off x Delay Per hand off

= 15 x 200

= 3000 msec

= 3 sec

iii) Failed handoff recovery delay

Recovery delay = 3 x 400

= 1200 ms

= 1.2 sec

iv) Additional recovery delay

Authentication = 350 + 250

= 600 ms

Total recovery delay = 2 × 600

= 1.2 sec

v) Overall interruption Delay

= Normal + Recovery + Remaining delay

= 3 + 1.2 + 1.2

= 5.4 sec

vi) Data affected during interruption

= Data rate × interruption delay

= 512 × 5.4

= 2764.8 Kbs

Convert to Kilobites

= 2764.8 / 8

= 345.6 KB

The total interruption time is 5.4 sec which exceeds the permitted delay by 2.4 seconds.

3)

i) Given:

Base transceiver stations (BTS) = 12

sectors = 3

Transceivers = 4 (TRX's)

Time slots = 8

Time slot per sector = 2

Subscribers = 1200

Calls per hour = 1.2

Average holding time = 90 sec

i) Traffic channel sector :-

$$TCH = (4 \times 8) - 2$$

$$= 32 - 2$$

$$= 30 \text{ traffic channel}$$

ii) Traffic channel per BTS and BSC:

$$BTS = 30 \times 3$$

$$= 90 \text{ TCH}$$

$$BSC: 90 \times 120$$

$$= 1080 \text{ TCH}$$

iii) offered traffic

$$A = (1200 \times 1.2 \times 90) / 3600$$

$$= 36 \text{ Erlangs}$$

iv) Channel utilization:

$$U = (36/30) \times 100$$

$$= 120\%$$

v) Present Abis bandwidth

= Traffic channel Bandwidth

= 90 x 16

= 1440 Kbps

= 1.44 Mbps

Percentage = (1.44/2.048) x 100

= 70.31 %

ii) Given:

One-way spectrum = 10 MHz

FDMA uses = 30 KHz

FDMA frequency reuse = N = 7

TDMA uses = 200 KHz

No. of users = 8 / carriers

TDMA reuse = N = 4

TDMA uses = 1.25 MHz

TDMA chip rate = 1.2288 Mcps

TDMA voice data rate = 9.6 Kbps

i) Total channels in FDMA:

$$C = \frac{\text{One-way spectrum}}{\text{FDMA}}$$

$$= \frac{10 \times 1000}{30}$$

= 333.3 channels

FDMA frequency reuse = 333 3/7

= 47.5

ii) Total channels in TDMA $= \frac{10 \times 1000}{200}$

$= 50$ channels

No. of users $= N = 8$

$= 50 \times 8$

$= 400$

FDMA frequency reuse $= 400/4$

$= 100$

iii) Rate $= \frac{\text{chip rate}}{\text{data rate}}$

$= \frac{1.2288}{9.6}$

$= 1.2$

Total channels $= \frac{1.25 \times 1000}{1.25}$

$= 1000$

4) Given:

Transfer files = 240

Each of size = 18 MB

Rate = 20 Kbps

Reduced overhead = 18%

Effective throughput = 85%

Packet retransmission = 12%

Each lasting = 2.5 sec

i) convert to megabits

$D = 240 \times 18$

$D = 4320$ MB

$D = 4320 \times 8$

$D = 34560$ Mb

ii) calculate aggregate GPRS

$R = 4 \times 20$

$R = 80$ KbPS

iii) After 18% overhead

$R_1 = 80 \times 0.82$

$R_1 = 65.6$ KbPS

After 85% efficiency.

$R_{effective} = 65.6 \times 0.85$

$R_e = 55.76$ KbPS

iv) calculate time

$$T = \frac{\text{Data Volume}}{\text{Through Put}}$$

$$T = \frac{38707.2 \times 1000}{55.76}$$

$T = 694175.04$ sec

mobility interruption time

$T_m = 8 \times 2.5$

$T_m = 20$ sec

5) Given:-

Software update size = 120 GB

No of mobiles = 2000

GSM rate = 9.6 Kbps

GPRS effective throughput = 64 Kbps

GPRS retransmission = 10%

5G rate = 25 Mbps

5G protocal overhead = 4%

i) Calculate volume

$D = 120 \times 1000$

$D = 120\,000$ MB

convert to megabits

$D = 120000 \times 8$

$D = 960000$ mb

ii) Calculate GSM

$R = 9.6$ Kbps

$T = \frac{960000 \times 1000}{9.6}$

$T = 100000000$ Sec

Into days $T = \frac{100000000}{86400}$

$T = 1157.41$ days

(iii) Calculate GPRS Time

R = 64 KBPS

Retransmission = 10%

D = 960000 × 1.10

D = 1056000 Mb

Transfer time

$$T = \frac{1056000 \times 1000}{64}$$

T = 16500000 sec

convert into days

$$T = \frac{16500000}{86400}$$

T = 190.97 days

(iv) Calculate 5G